



WEEK 1

SOFTWARE DEVELOPMENT TOOLS AND ENVIRONMENTS



Presented by **Asst. Prof. Dr. Tuchsanaï Ploysuwan**



- Git was developed in 2005 by **Linus Torvalds**
- Git is **Version control system** is a system that records changes to a file or set file over time so that you. can restore specific version later
- Git is a **Distributed Version Control System**







Git – What and Why

Oh boy, I sure do
love playing my
video games!



I'm going to save
my game now in
case I die soon!



Oh jeez, this is
going to be a
difficult fight!



ughhhhhh I died!





Thank heavens I
saved my game! I
can just revert!



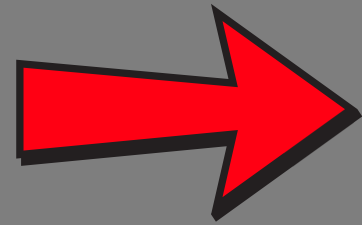
Before Version Control System



Before Version Control System



Before Version Control System



Before Version Control System



Version 1.0



Version 2.0

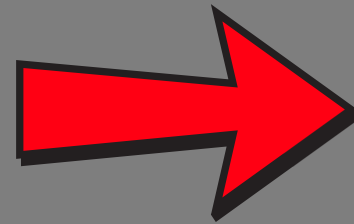


I have a great idea,
send me your code



running code version 1.0

Before Version Control System



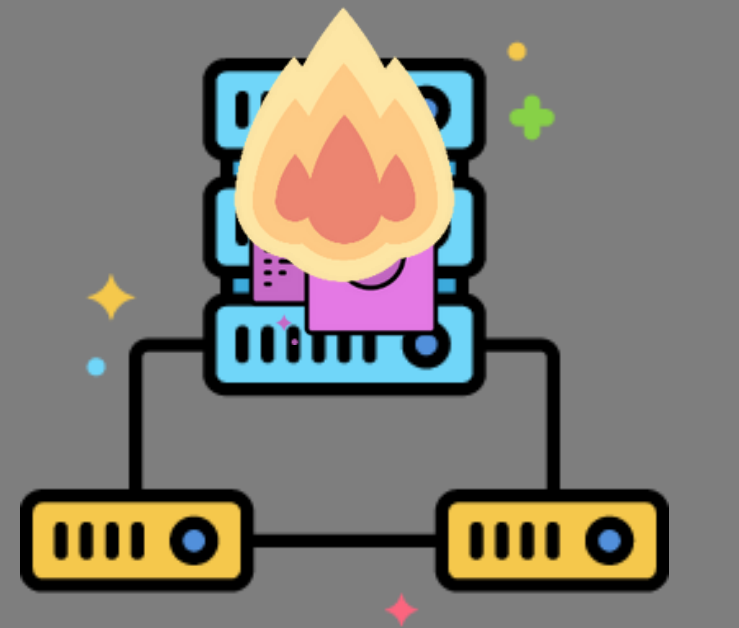
Before Version Control System



Before Version Control System

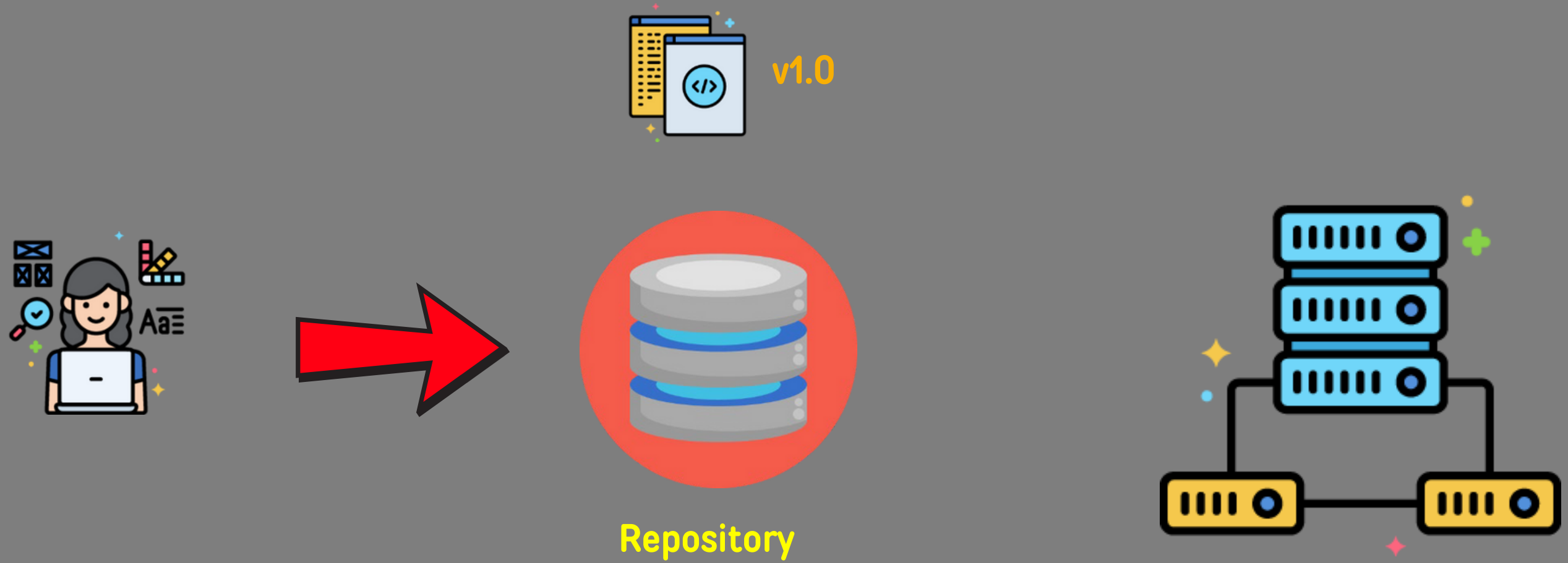


Version 2.0

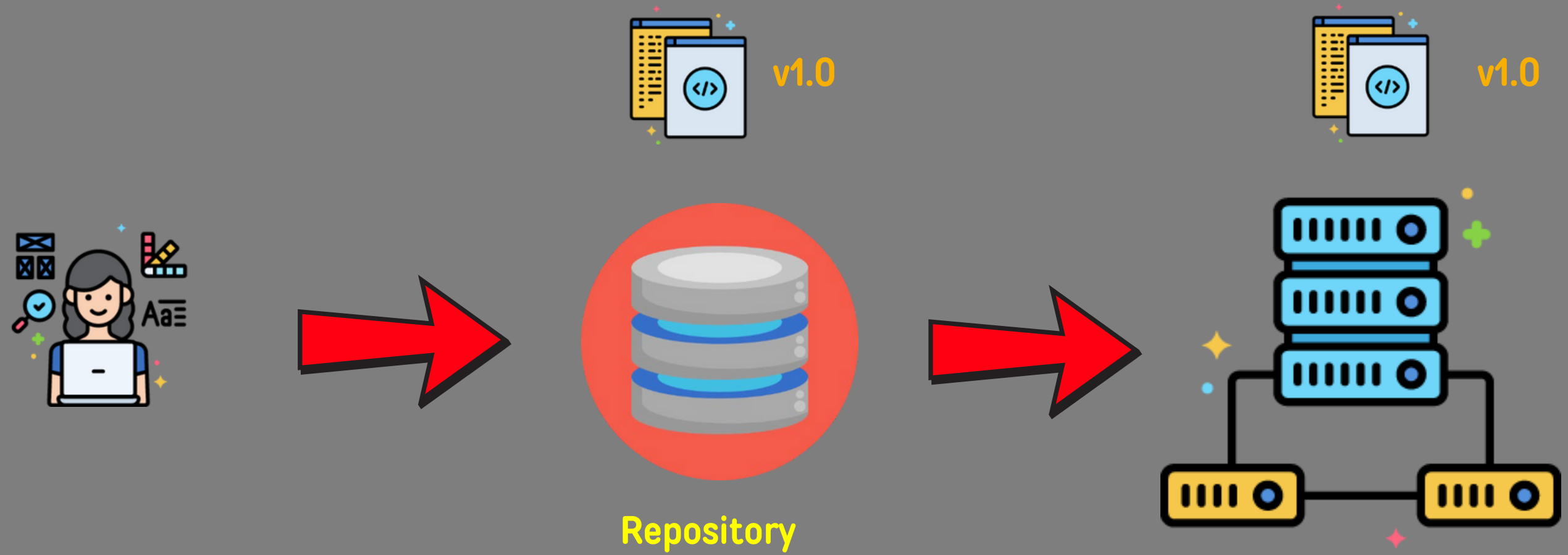


- Rollback is time consuming
- No audit tracking
- Not scalable for large teams

Version Control System



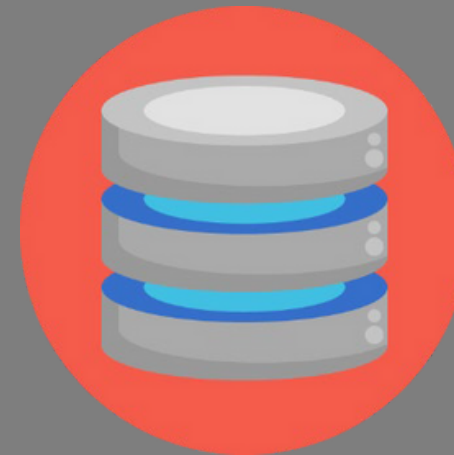
Version Control System



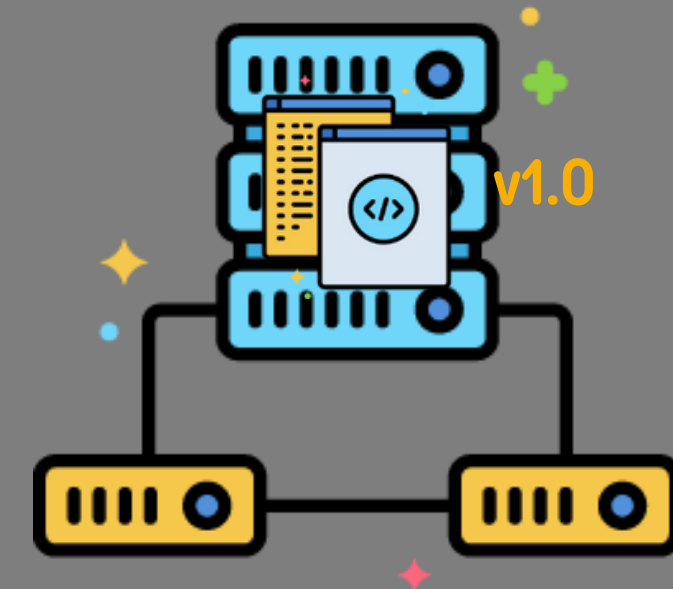
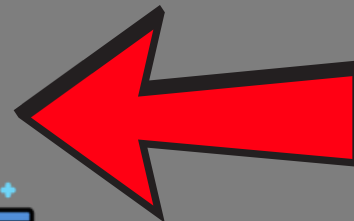
Version Control System



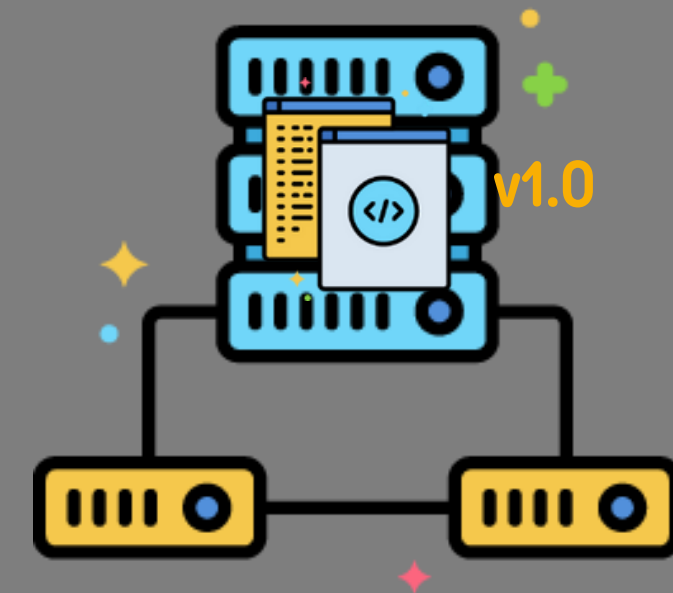
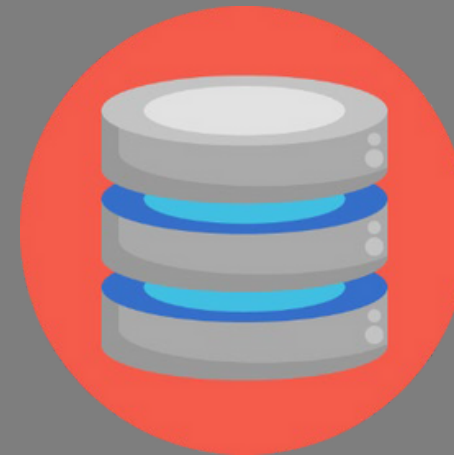
Fetch the code



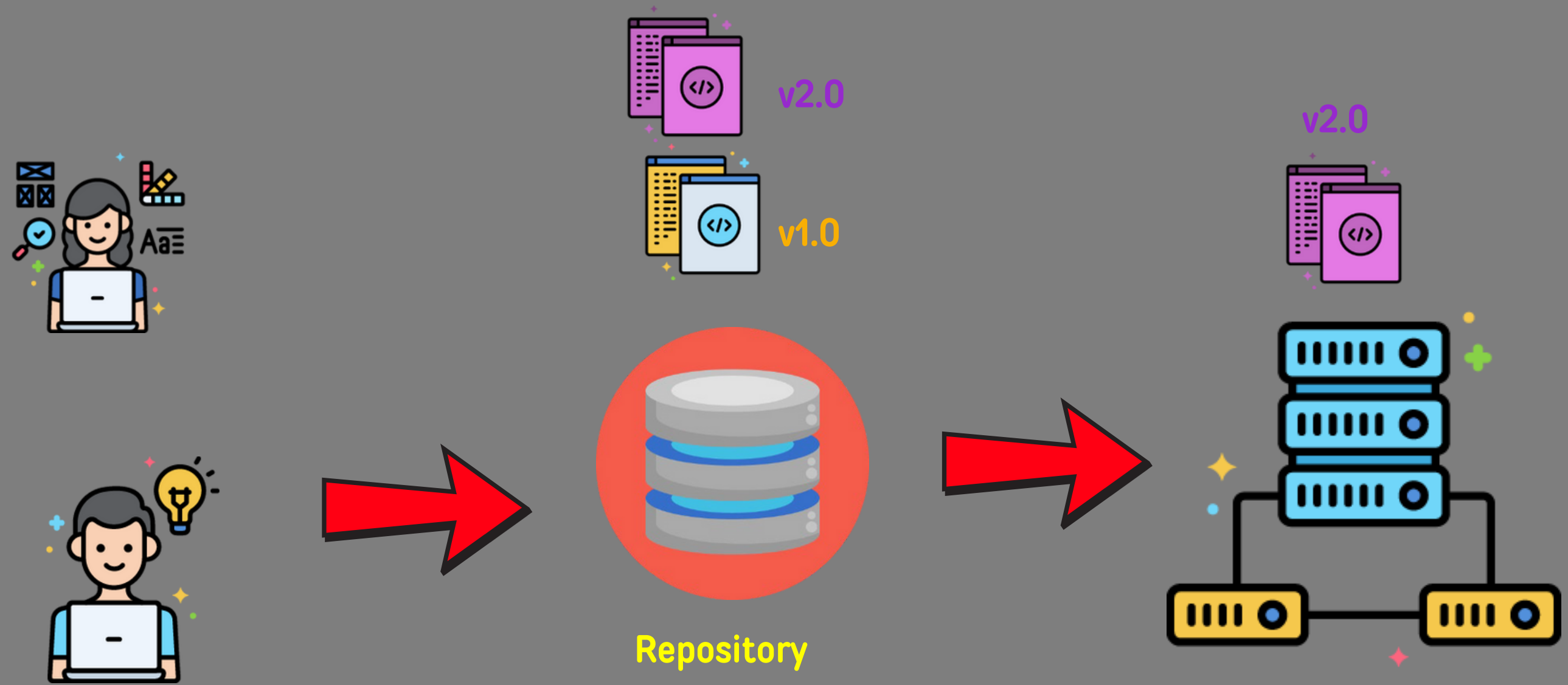
v1.0
Repository



Version Control System



Version Control System



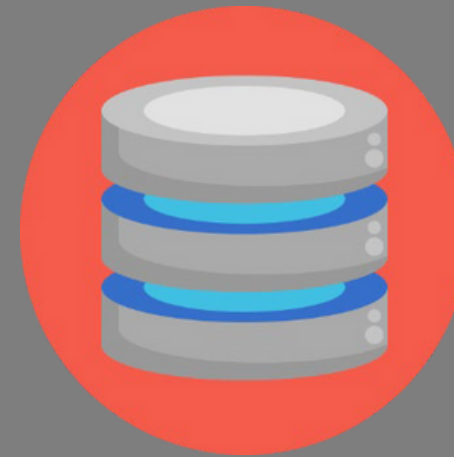
Version Control System



Repository



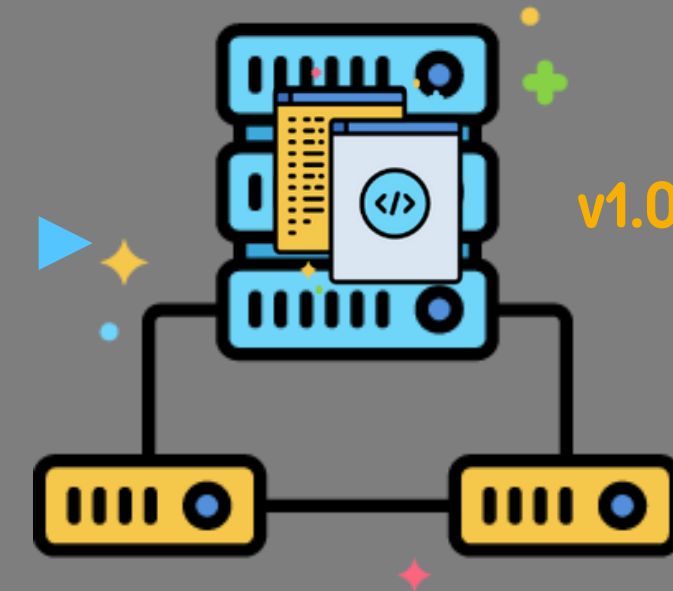
Version Control System



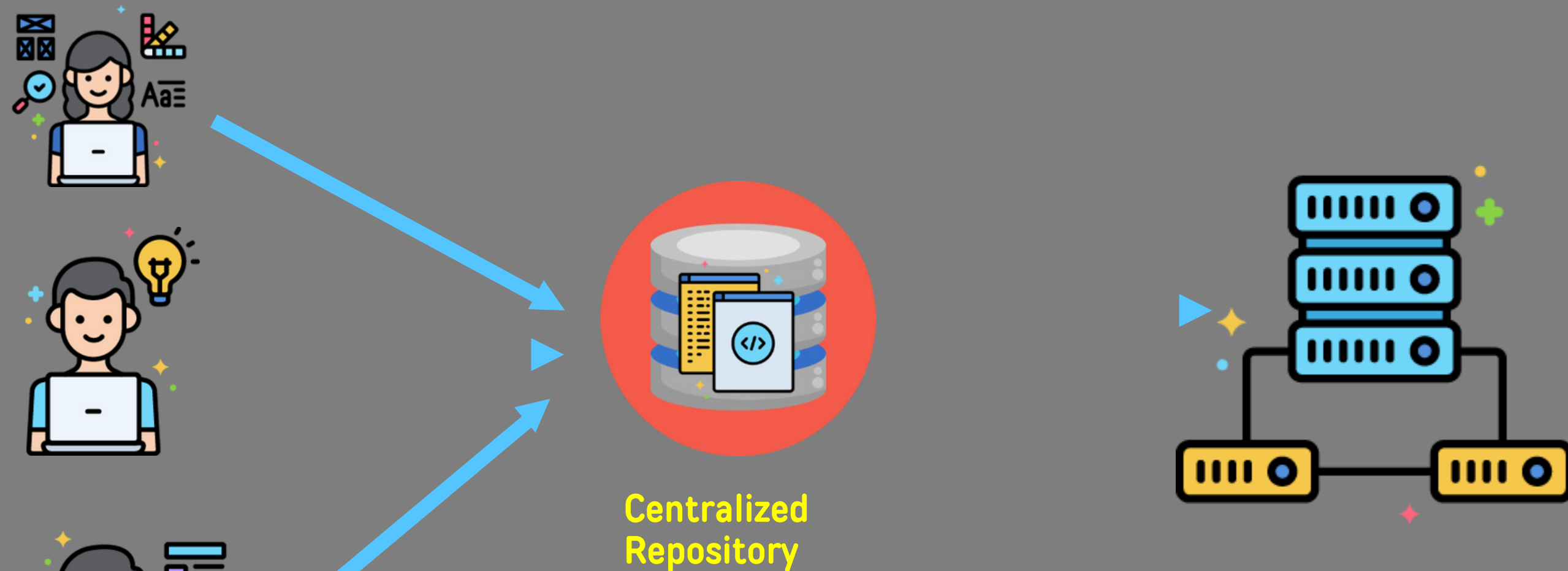
Repository

Why Git?

- Distributed

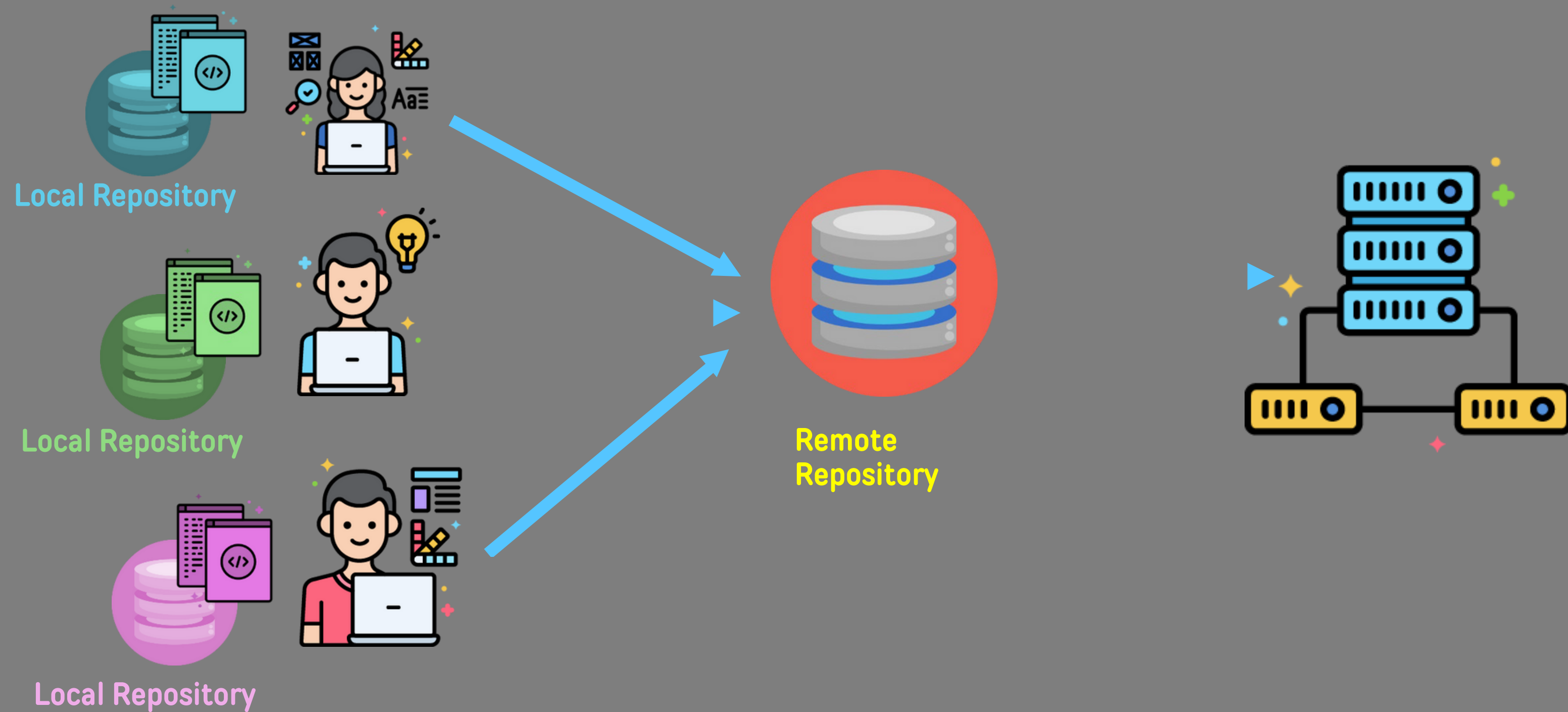


Centralized Version Control System



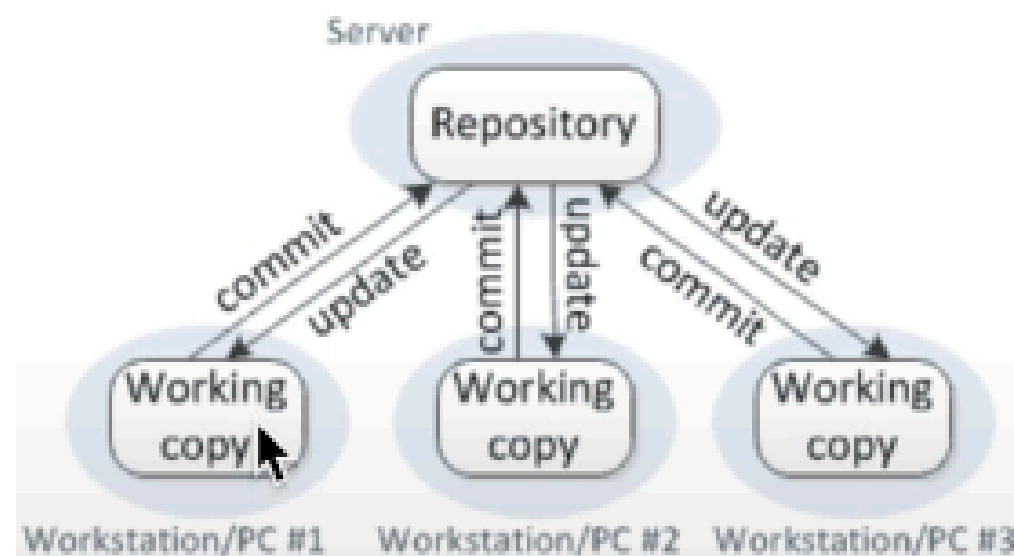
- Single point of failure
- Requires constant connectivity
- E.G – Subversion, Endeavor

Distributed Version Control System

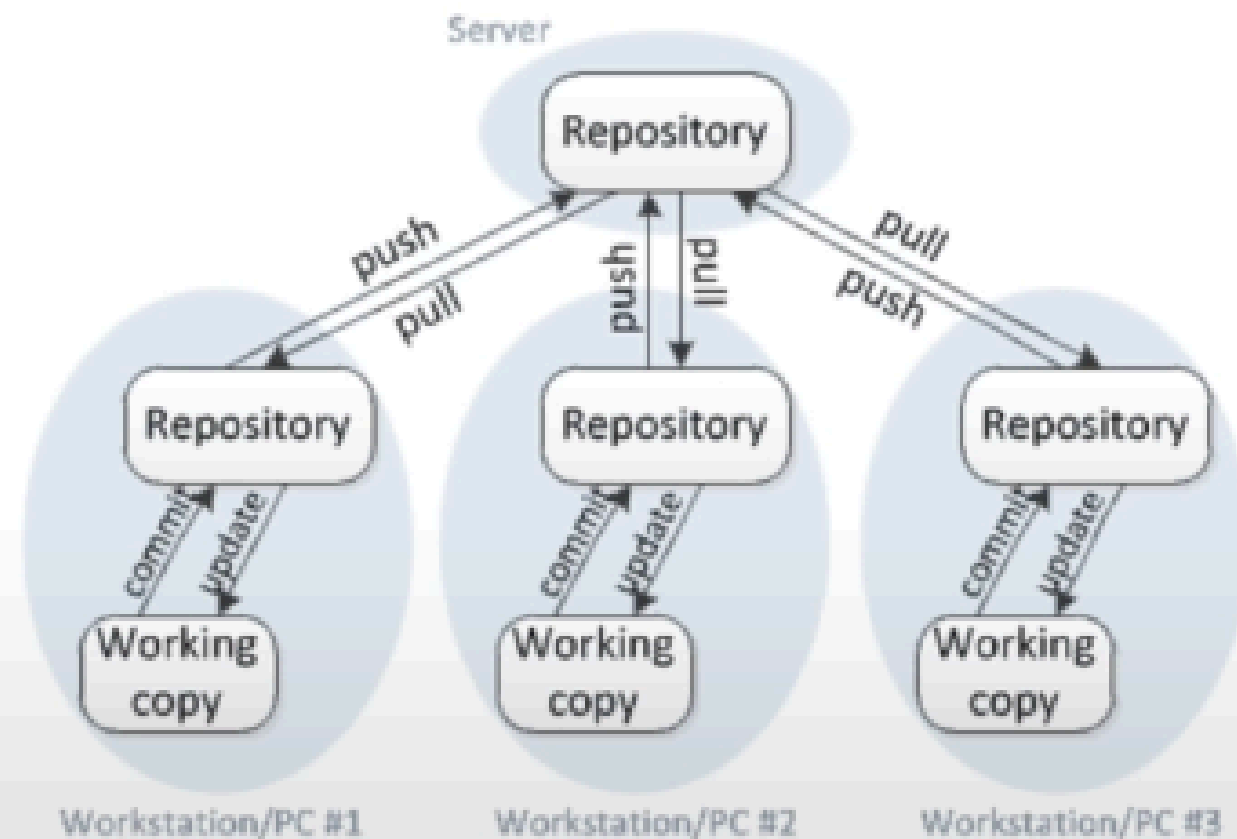


Centralized vs **Distributed** Version Control

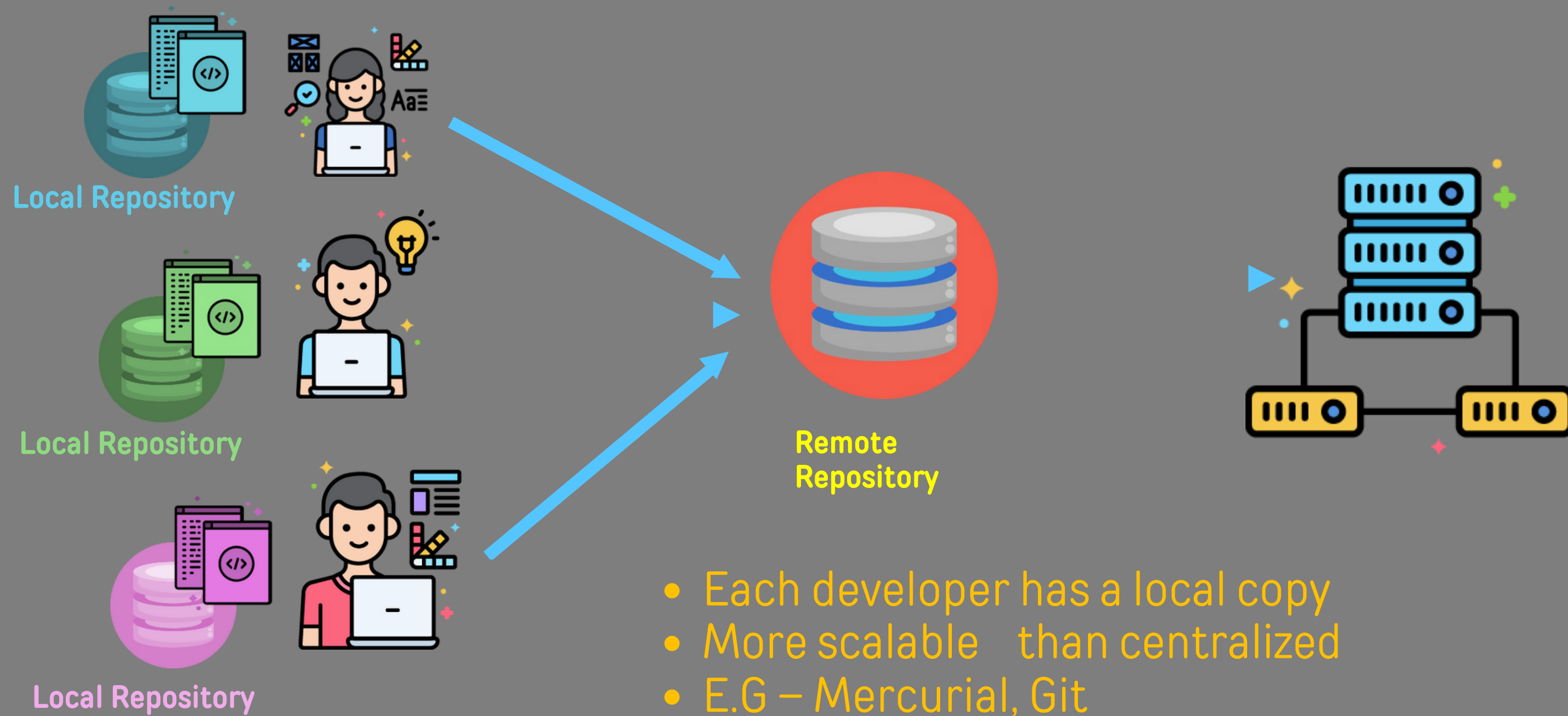
Centralized version control



Distributed version control



Distributed Version Control System



Version Control System



Why Git?

- Distributed
- Performant
- Detailed audit tracking
- Open source
 - Free!
 - Implemented with Kubernetes GitOps, integration with Jenkins and other DevOps tools
 - GitHub, GitLab, Code Commit are all based on Git

Git vs GitHub

Git Vs. GitHub

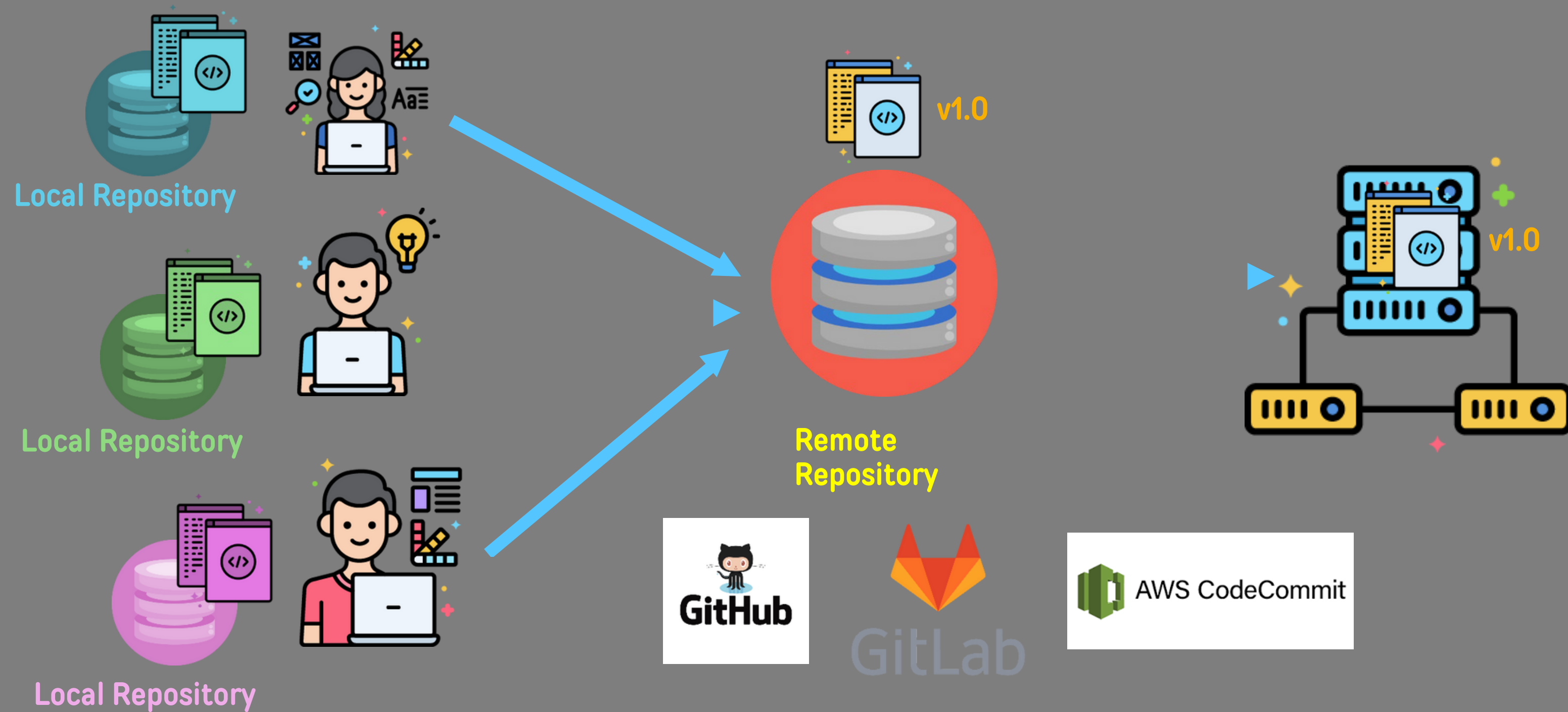


- Version Control System
- Installed locally on the system
- Created in 2005, by Linus Torvalds
- Open source, and used in multiple cloud repository services



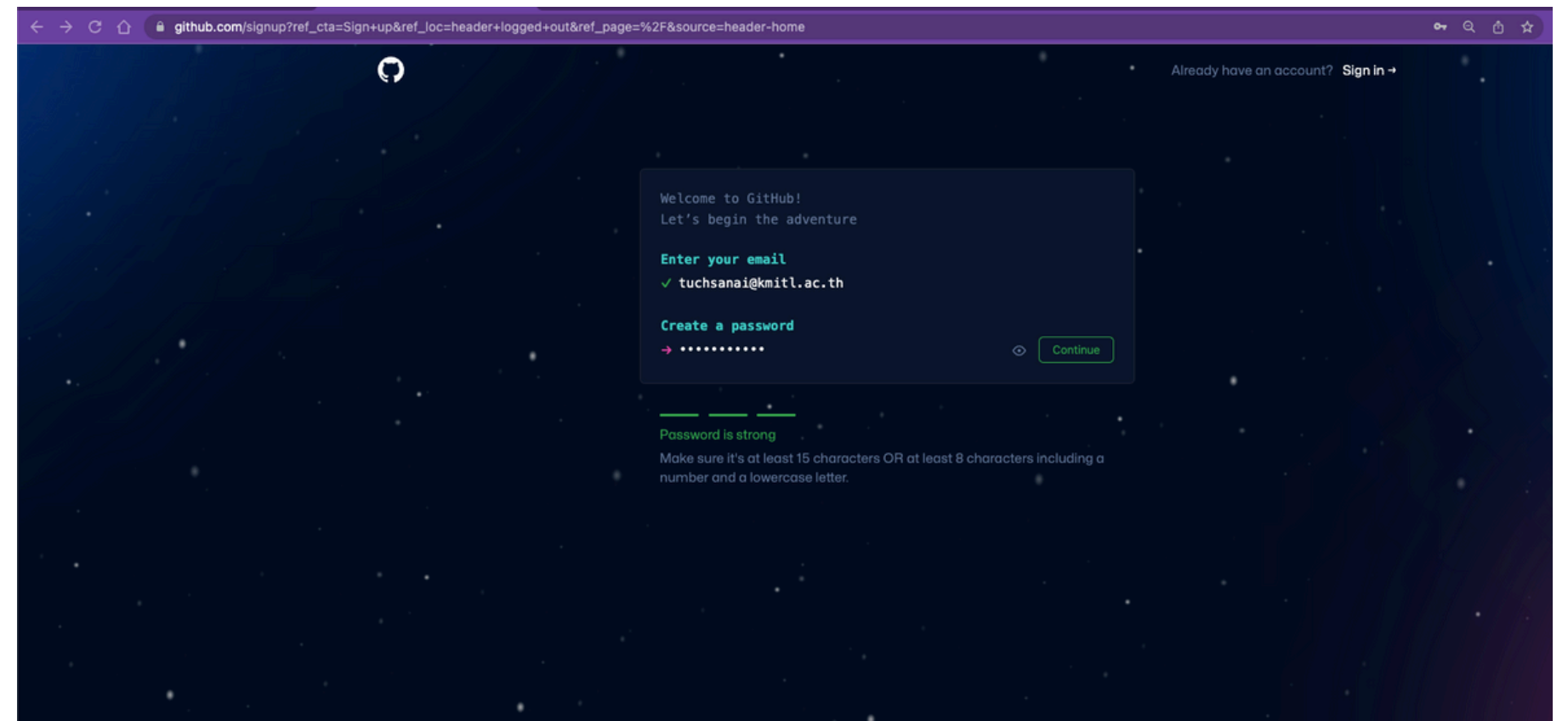
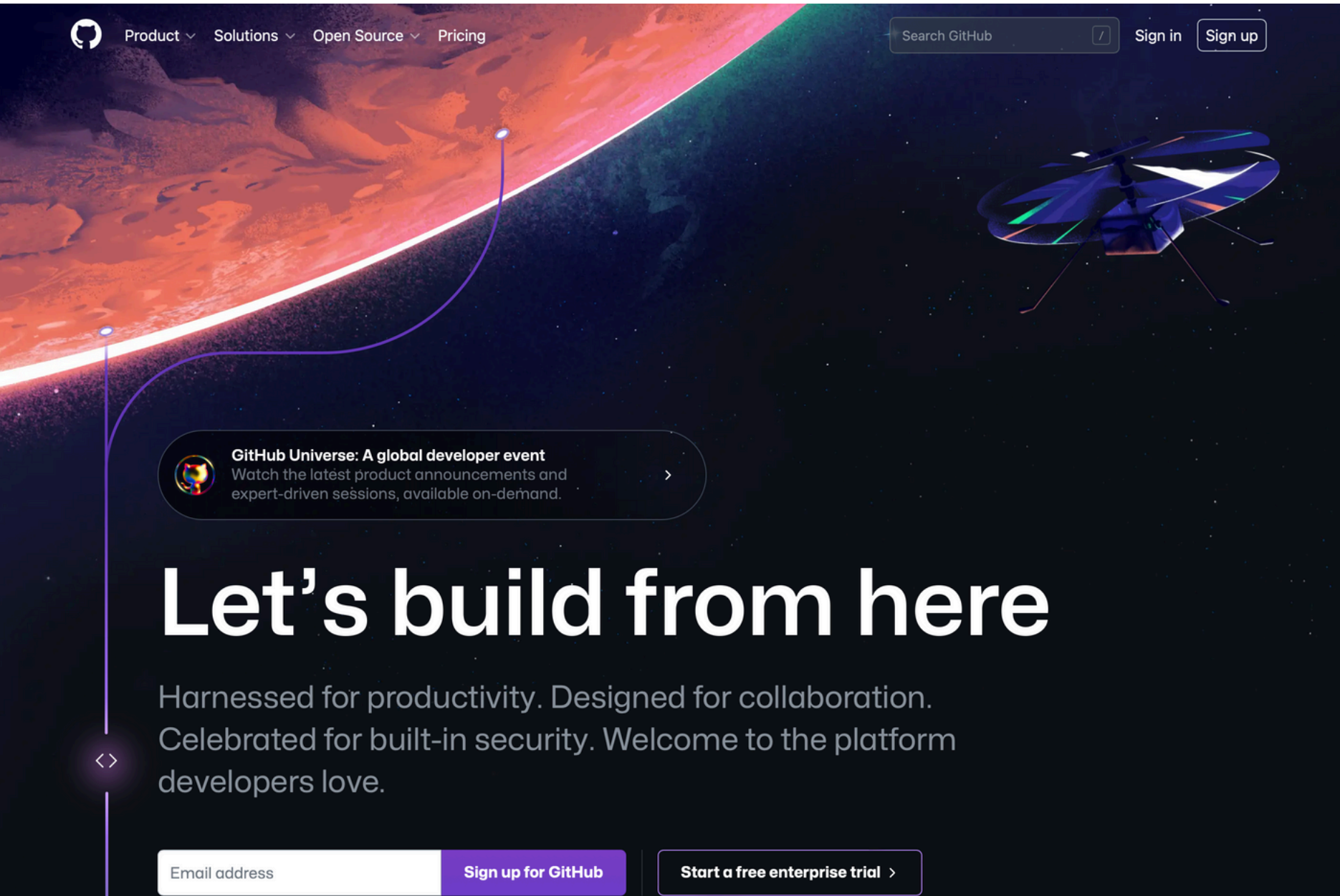
- Git repository hosting services with other features
- Runs on the cloud
- Created in 2008, currently owned by Microsoft
- Not open source, have free and paid tiers

Distributed Version Control System



Week 1 - Starting with Git

Sign Up : <https://github.com>



<https://github.com>

Installing Git

Week 1 - Starting with Git

- **MacOS or Linux Users:**
 - Congrats! You already have Git installed on your machine since it comes pre-installed as part of your OS.
 - To confirm this, open up a terminal and type:
 - **git --version**
 - **>> git version 2.25.1 (Apple Git-128)**

Week 1 - Starting with Git

- **MacOS or Linux Users:**
 - If you wish to update or re-install git, you can do this by simply selecting the MacOS or Linux links on the official git website:
 - **<https://git-scm.com/downloads>**

Week 1 - Starting with Git

- **MacOS or Linux Users:**

- Our suggested text editor for this course is VS Code:
 - <https://code.visualstudio.com/>
- Its created by Microsoft and has direct integrations with GitHub and is one of the most popular text editors today.
- You can follow along with any text editor you prefer however.



Week 1 - Starting with Git

- **Windows Users:**

- Our *HIGHLY* recommend text editor for this course is VS Code:
 - <https://code.visualstudio.com/>
- Why *HIGHLY* recommended?
 - Windows + VS Code + GitHub
 - Upon installing git you will be asked to select a default editor, you'll need VS Code installed to select it as default



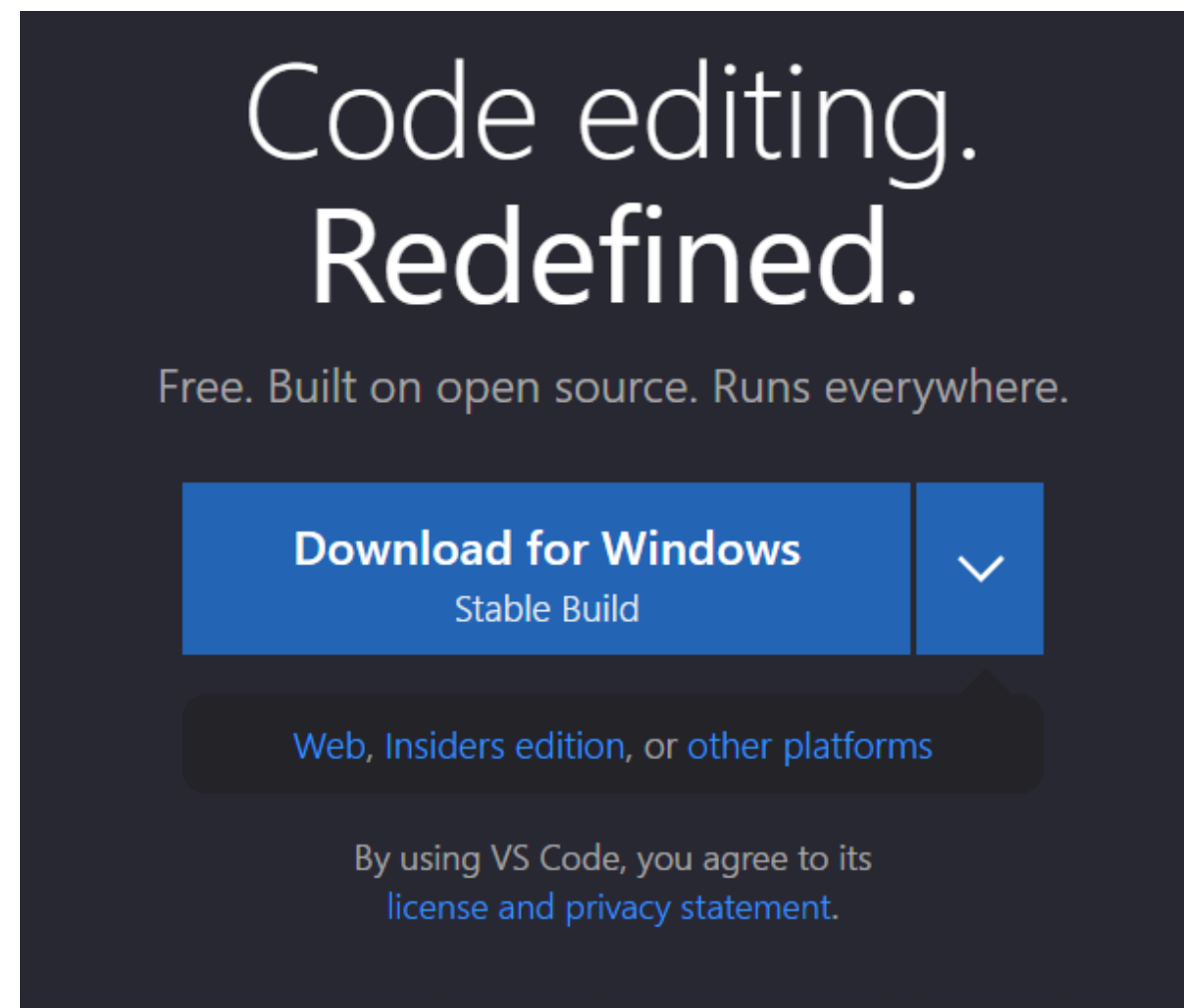
Week 1 - Starting with Git

- **Windows Users:**

- Go to:

- <https://code.visualstudio.com/>

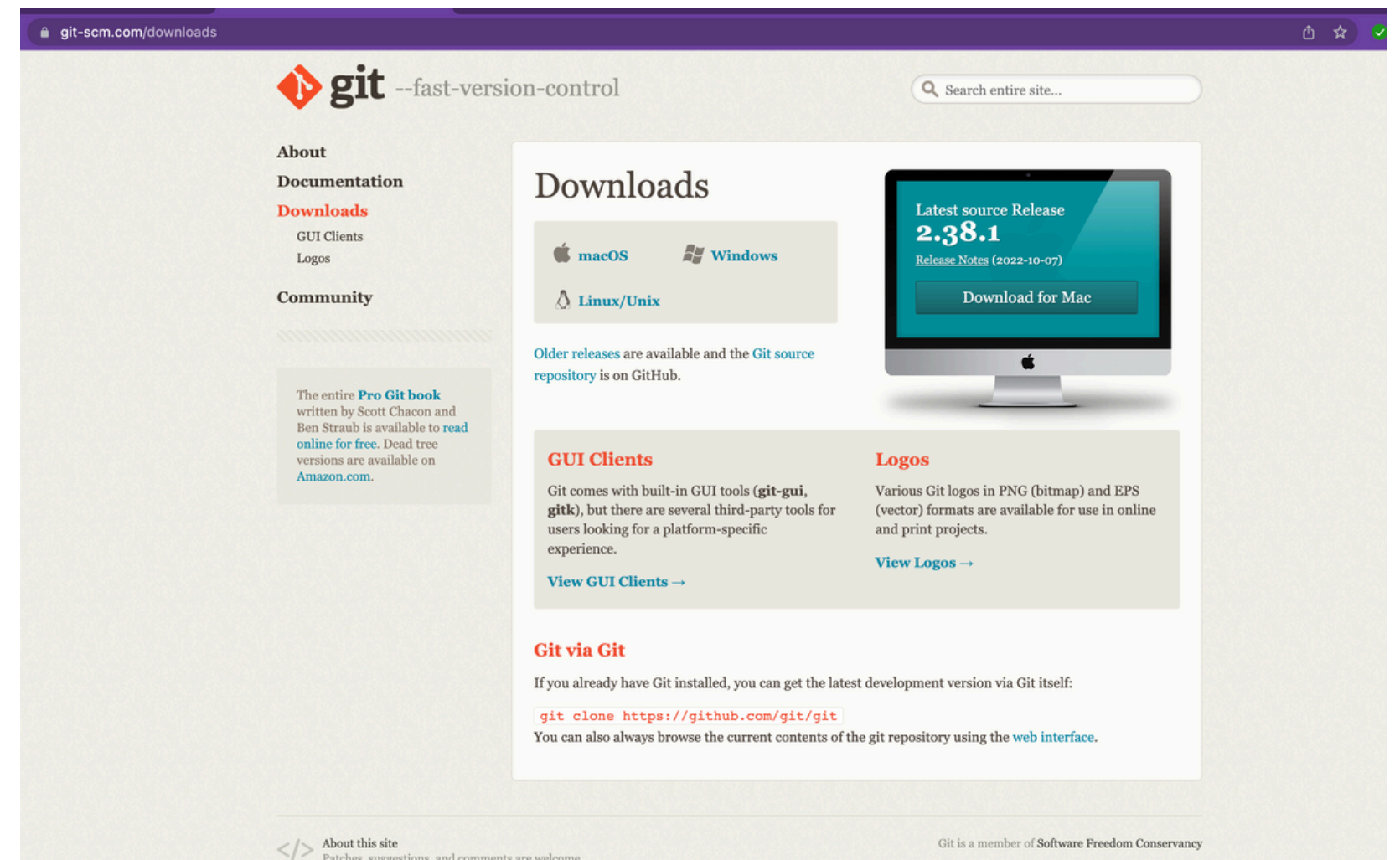
- Download with Default Settings:





Week 1 - Starting with Git

- **Windows Users:**
 - Next we'll download git, go to:
 - <https://git-scm.com/>



DAY 1

Configure Git

Week 1 - Starting with Git

- You can check the current configuration with the commands:
- The configuration commands will be:
 - **git config --global user.name "user"**
 - **git config --global user.email "email"**
- If switch with another github account
 - **git config --global user.name "user"**
 - **git config --global user.email "email"**
 - **git config --global credential.username "user"**

- -- ลบ config เก่า --
 - git config --global --unset user.name
 - git config --global --unset user.email
 - git config --global --unset credential.username
-
- -- ลบ origin คนเก่า ถ้า commit ไม่ได้ --
 - git remote remove origin

Show global Git configuration?

```
git config --list or git config -l
```

or look at your `~/.gitconfig` file. The local configuration will be in your repository's `.git/config` file.

```
git config --list --show-origin
```

Week 1 - Starting with Git

- Let's head over to our command line interface to set-up our Git configuration:
 - Git Bash
 - Terminal
 - Command Prompt

DAY 1

Creating a Git Repository

Day 1 - Starting with Git

- How can we create a Git Repository?
 - **git init**
 - This command initializes a Git Repository on your local machine.
 - You only need to run this command once per project.
 - **git status**
 - This command will report back the status of your Git repository.

Day 1 - Starting with Git

- How can we create a Git Repository?
 - Upon creating a repository with **git init** you will create a hidden .git file.
 - The .git file is a hidden file that manages the versioning of the files inside the Git repository.

Day 1 - Starting with Git

- Git inside a Folder/Directory:
 - Upon creating a Git Repository, all the folders/directories inside the top level Git Repository will also be part of that Repository, meaning all the changes are tracked.

Day 1 - Starting with Git

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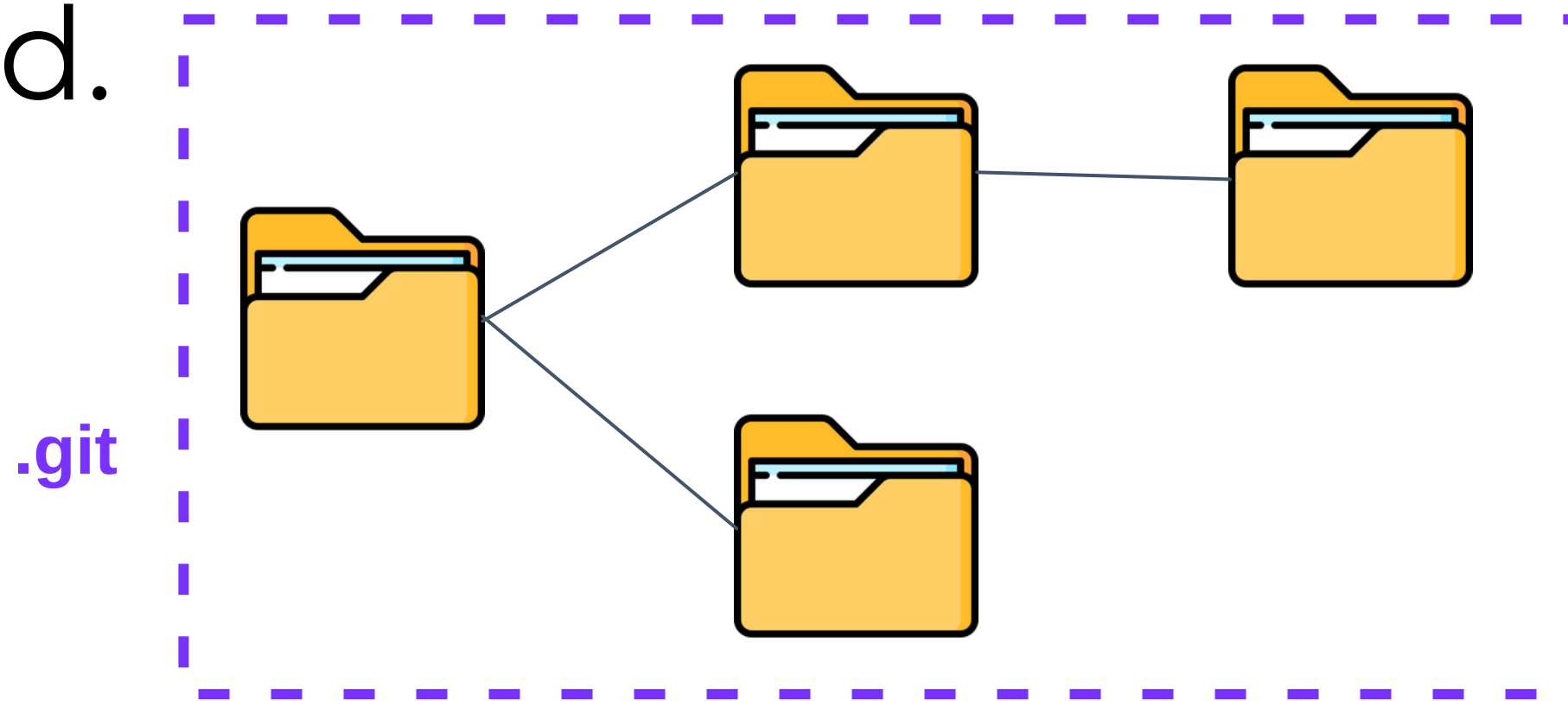
Day 1 - Starting with Git

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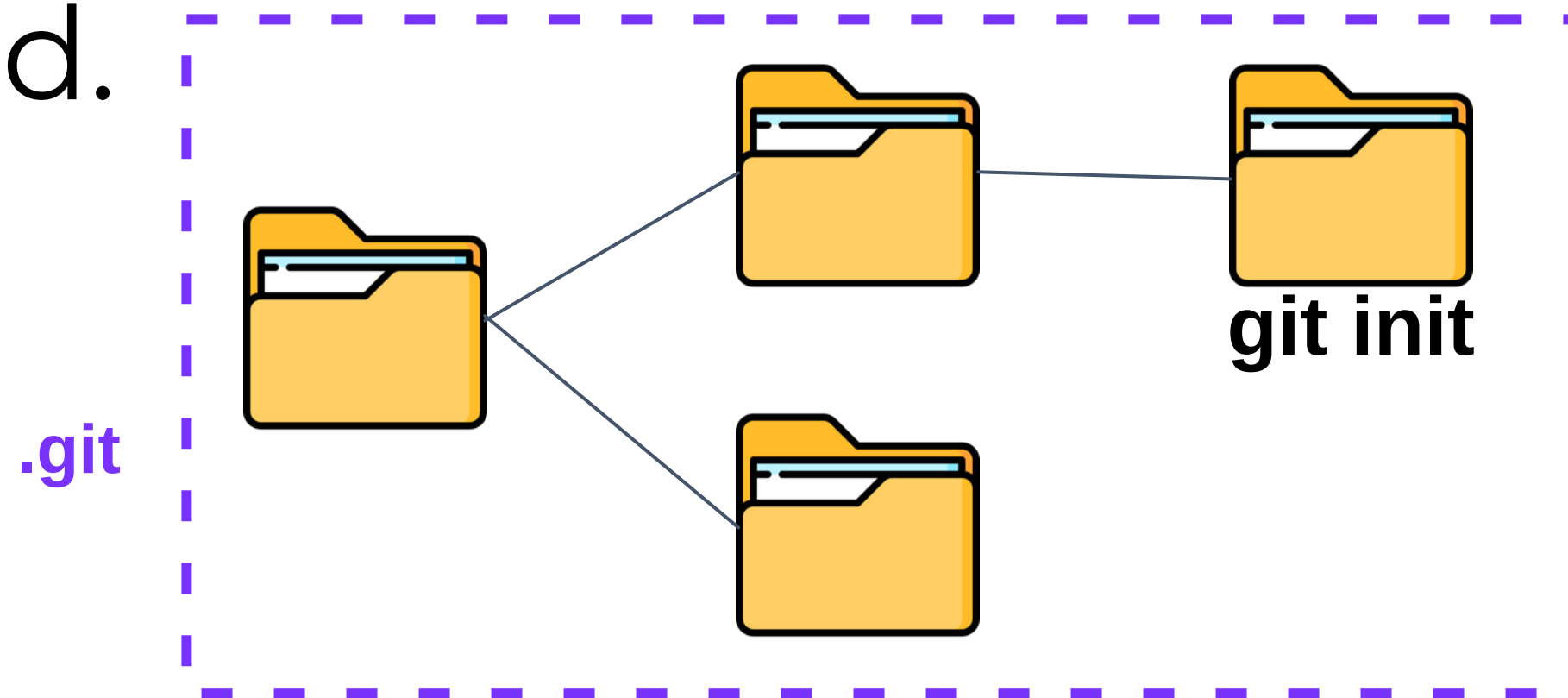
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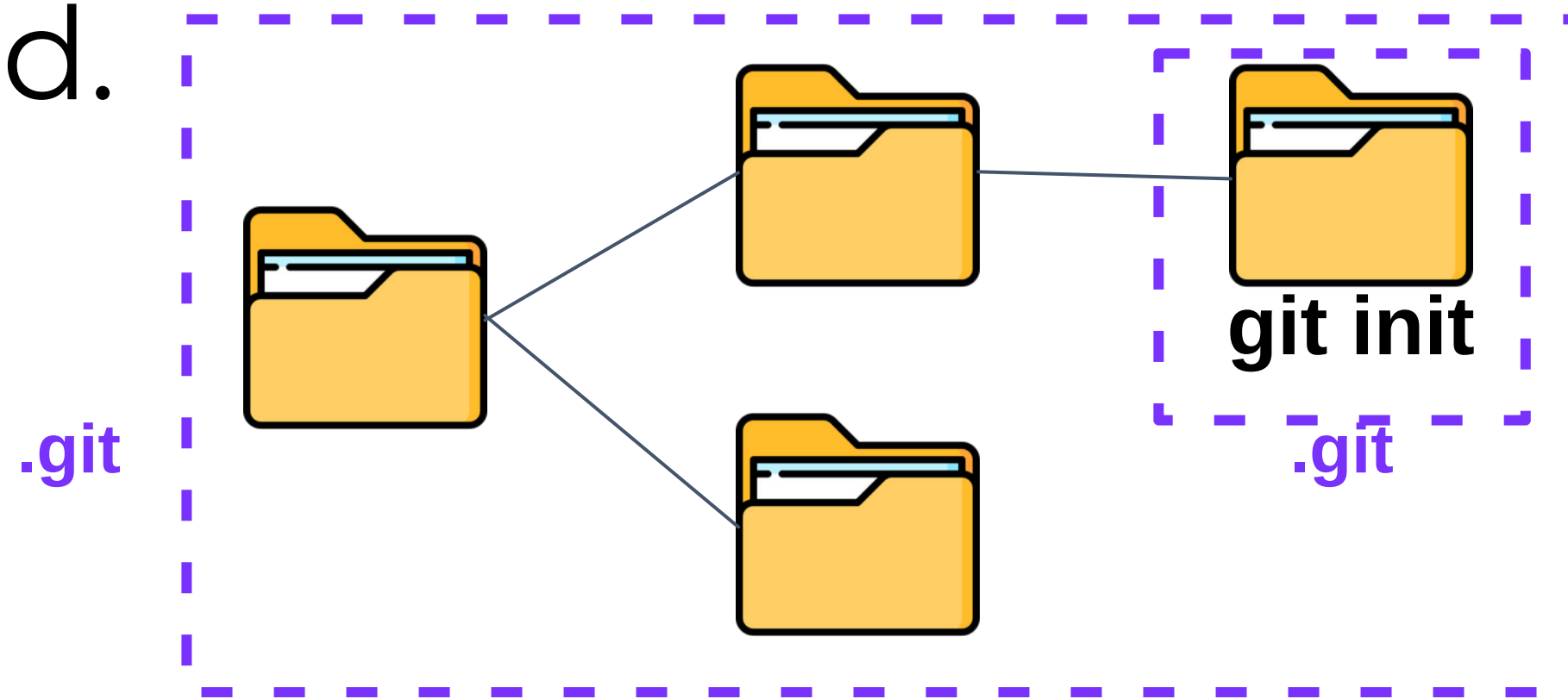
Day 1 - Starting with Git

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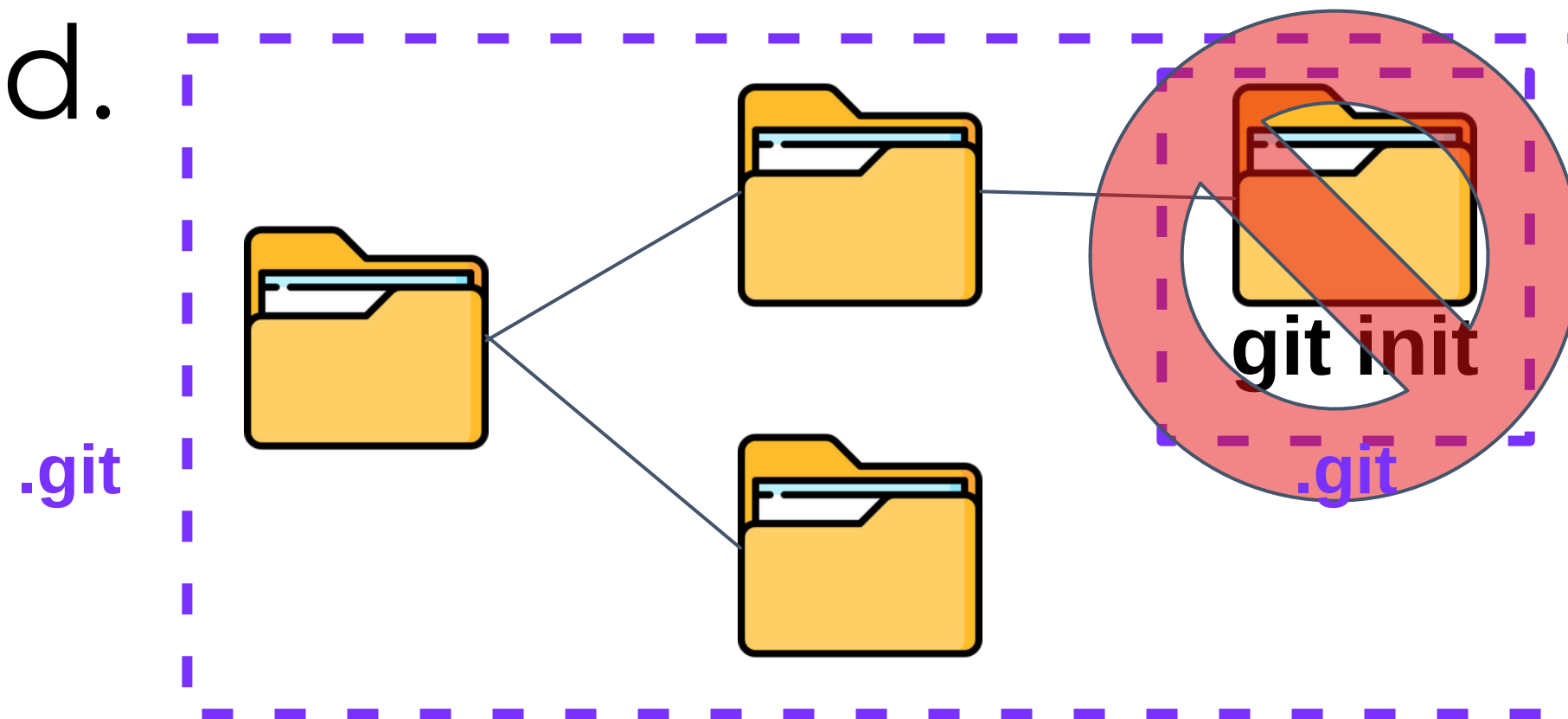
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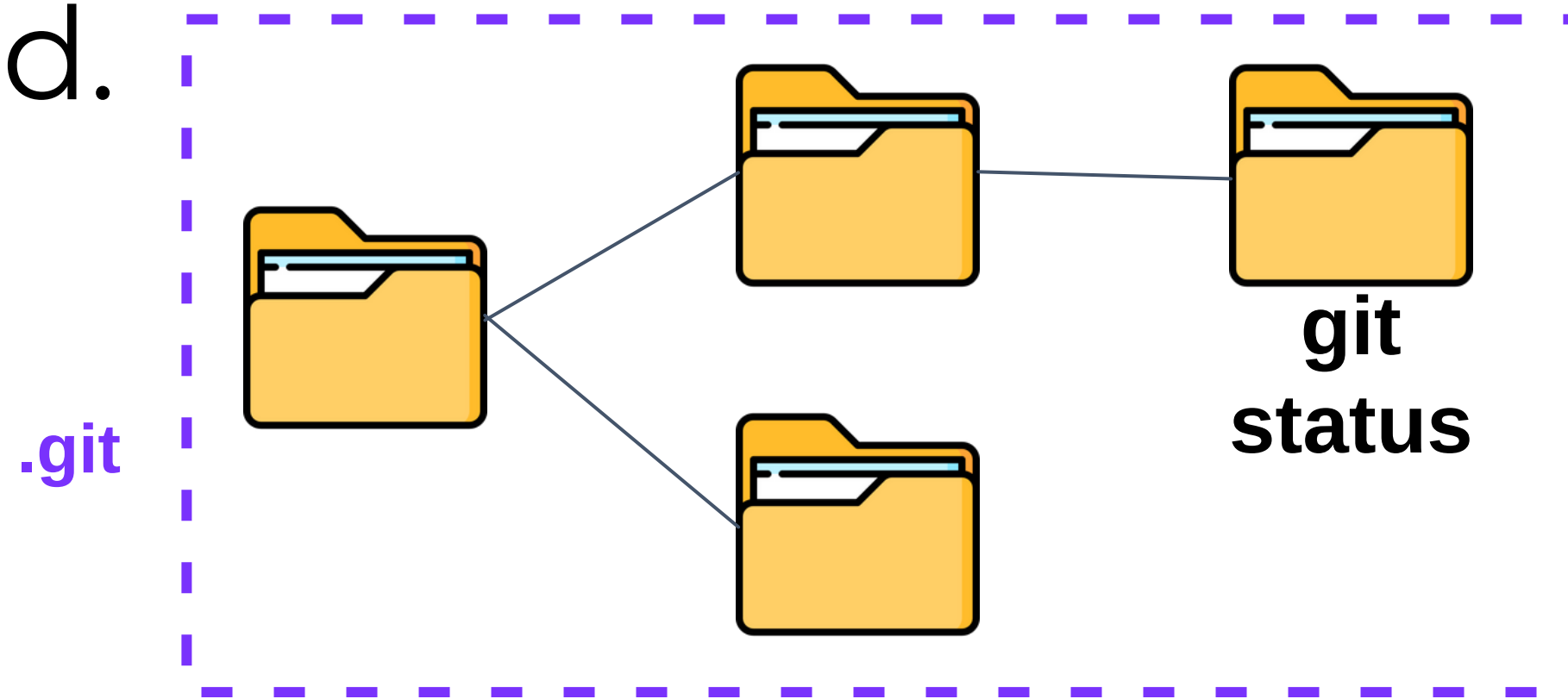
Day 1 - Starting with Git

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Day 1 - Starting with Git

- Git inside a Folder/Directory:
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Ignoring Files

We can tell Git which files and directories to ignore in a given repository, using a `.gitignore` file. This is useful for files you know you **NEVER** want to commit, including:

- Secrets, API keys, credentials, etc.
- Operating System files (`.DS_Store` on Mac)
- Log files
- Dependencies & packages



.gitignore

Create a file called `.gitignore` in the root of a repository. Inside the file, we can write patterns to tell Git which files & folders to ignore:

`.DS_Store` will ignore files named `.DS_Store`

`folderName/` will ignore an entire directory

`*.log` will ignore any files with the `.log` extension

<https://www.toptal.com/developers/gitignore>



gitignore.io

สร้างไฟล์ .gitignore ที่มีประโยชน์สำหรับโปรเจกต์ของคุณ

ค้นหาแบบปฏิบัติการ IDE หรือภาษาการเขียนโปรแกรม

สร้าง

ชอร์สโค้ด | ลองอ่านเอกสาร คำสั่งของ คอมมิตไลน์ ดูลิ!



DAY 1

Private Repositories and Tokens

Day 1 - Starting with Git

- Clone Syntax with PAT:

git clone https://username:YOUR_TOKEN@github.com/username/repo.git

- Previously we used:

git clone https://github.com/account/repo.git

Create a Personal Access Token

Q Type / to search

>_

+ ▾

🔄

🔗

📧



You unlocked new Achievements with private contributions! Show them off by including private contributions in your Profile in [settings](#).

Pinned

Public

⋮

Jupyter Notebook ⭐ 1 🍷 7

⋮

Jupyter Notebook ⭐ 10 🍷 14

⋮

Jupyter Notebook ⭐ 1 🍷 1

Customize your pins

746 contributions in the last year

Contribution settings ▾

	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Mon	■		■		■		■	■	■	■	■	■	■
Wed		■	■	■	■	■	■	■	■	■	■	■	■
Fri	■	■	■	■	■	■	■	■	■	■	■	■	■

1

Click on your profile picture in the upper-right corner of the screen and select "Settings."

↑

😊 Set status

👤 Your profile

📁 Your repositories

📁 Your projects

📁 Your codespaces

📁 Your organizations

🌐 Your enterprises

⭐ Your stars

💖 Your sponsors

📄 Your gists

📈 Upgrade

🌐 Try Enterprise

👤 Try Copilot

🧪 Feature preview

⚙️ Settings

📖 GitHub Docs

👤 GitHub Support

Sign out

2

Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

ss

Billing and plans

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Moderation

, planning, and automation

Repositories

Codespaces

Packages

Copilot

Pages

Saved replies

Privacy

Code security and analysis

Integrations

Applications

Scheduled reminders

Invites

Security log

Sponsorship log

Developer settings

Public profile

Name

tuchsanai

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You have set your email address to private. To toggle email privacy, go to [email settings](#) and uncheck "Keep my email address private."

Bio

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

Pronouns

Don't specify

URL

Social accounts

Link to social profile

Link to social profile

Link to social profile

Link to social profile

Company

You can @mention your company's GitHub organization to link to it.

Location

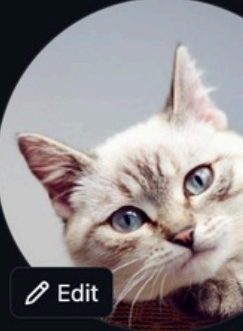
☐ Display current local time

Other users will see the time difference from their local time.

the fields on this page are optional and can be deleted at any time, and by filling them out, you're giving us consent to share this data wherever your user profile appears. Please see our [privacy statement](#) to learn more about how we use this information.

Update profile

Profile picture



Edit

In the left sidebar, select "Developer settings."

Settings / Developer Settings

Type to search

GitHub Apps

OAuth Apps

Personal access tokens

GitHub Apps

New GitHub App

Want to build something that integrates with and extends GitHub? [Register a new GitHub App](#) to get started developing on the GitHub API. You can also read more about building GitHub Apps in our [developer documentation](#).

4

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Settings / Developer Settings

Type to search

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

Personal access tokens (classic)

Generate new token

Need an API token for scripts or testing? [Generate a personal access token](#) for quick access to the GitHub API.

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

5

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6

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

test

What's this token for?

Expiration *

7 days

The token will expire on Tue, Sep 26 2023

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events

☒ repo

repo:status

repo_deployment

public_repo

repo:invite

security_events

Full control of private repositories

Access commit status

Access deployment status

Access public repositories

Access repository invitations

Read and write security events

☐ workflow

Update GitHub Action workflows

☐ write:packages

☐ read:packages

Upload packages to GitHub Package Registry

Download packages from GitHub Package Registry

☐ delete:packages

Delete packages from GitHub Package Registry

☐ admin:org

☐ write:org

☐ read:org

☐ manage_runners:org

Full control of orgs and teams, read and write org projects

Read and write org and team membership, read and write org projects

Read org and team membership, read org projects

Manage org runners and runner groups

☐ admin:public_key

☐ write:public_key

☐ read:public_key

Full control of user public keys

Write user public keys

Read user public keys

☒ admin:repo_hook

☒ write:repo_hook

☒ read:repo_hook

Full control of repository hooks

Write repository hooks

Read repository hooks

☐ admin:org_hook

Full control of organization hooks

☐ gist

Create gists

☐ notifications

Access notifications

☐ user

☐ read:user

☐ user:email

☐ user:follow

Update ALL user data

Read ALL user profile data

Access user email addresses (read-only)

Follow and unfollow users

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Beta

Tokens (classic)

Personal access tokens (classic)

Generate new token

Revoke all

Tokens you have generated that can be used to access the [GitHub API](#).

Make sure to copy your personal access token now. You won't be able to see it again!

✓ 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

Delete

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

DAY 1

Exercise

Day 1 - Starting with Git

- **Exercise Tasks:**

- Create a new **Private Repository** on GitHub.
- Initialize your repository with README, license and gitignore.
- create new test.txt with random data
- Clone your Repository using the Command Line .

github.com/dashboard

Dashboard

Type **Z** to search

Top repositories

New

Find a repository...

Join GitHub Education!

GitHub Education opens doors to new skills, tools, and a collaborative community eager to drive innovation. Join us and build a foundation for your future in technology.

Free and discounted services for teachers and students.

Copilot

Turn natural language prompts into coding suggestions

Heroku

Build, run, and operate applications entirely in the cloud.

Microsoft Azure

Access to Microsoft Azure cloud services and learning resources

Join GitHub Education

Home

Give feedback

Filter

II Sourcell

made this repository public

12 hours ago

II Sourcell/DualStrike_AI_For_Lymphoma

Python

1

Star

ALucek

made this repository public

20 hours ago

ALucek/chunking-strategies

Jupyter Notebook

Star

github.com/dashboard

Dashboard

Type **Z** to search

Top repositories

New

Find a repository...

Tuchsanai/DevTools

Tuchsanai/AIMaster-seagate-training-2024

Tuchsanai/DL-FOR-COMPUTER-VISION

Tuchsanai/DGX_2024

Tuchsanai/tp_idea1_2024

Tuchsanai/Machine_Learning

Tuchsanai/pytorch_docker

Show more

Join GitHub Education!

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Free and discounted services for teachers and students.

Join GitHub Education

Home

II Sourcell

made this repository public

12 hours ago

II Sourcell/DualStrike_AI_For_Lymphoma

Python

1

ALucek

made this repository public

20 hours ago

ALucek/chunking-strategies

Jupyter Notebook

Tuchsanai

tuchsanai

Set status

Your profile

Your repositories

Your Copilot

Your projects

Your stars

Your gists

Your organizations

Your enterprises

Your sponsors

Try Enterprise

Feature preview

Settings

GitHub Docs

GitHub Support

GitHub Community

Sign out

Click New repository

The screenshot shows the GitHub profile page for user 'tuchsanai'. The 'Repositories' tab is selected, showing a list of repositories. A large orange rectangle covers the bottom half of the page. A red circle highlights the '+ New repository' button in the top right corner. A red number '3' is placed next to the button.

github.com/Tuchsanai?tab=repositories

Tuchsanai

Overview Repositories 125 Projects Packages Stars 13

Find a repository... Type Language

MLOps_Class Public

06026241 MACHINE LEARNING OPERATIONS

Jupyter Notebook 1 Updated yesterday

tuchsanai Tuchsanai

Edit profile

28 followers · 34 following

Achievements

New repository

3

The screenshot shows the 'Create a new repository' page. The 'Repository name' field is filled with 'test_week1'. The 'Owner' is 'Tuchsanai'. The 'Description' field is empty. The 'Public' radio button is selected. The 'Add a README file' checkbox is checked. The '.gitignore template: Python' dropdown is selected. The 'License: None' dropdown is selected. A red circle highlights the 'Create repository' button at the bottom right. A red number '4' is placed next to the 'Repository name' field. A red number '5' is placed next to the 'Public' radio button. A red number '6' is placed next to the 'License: None' dropdown.

github.com/new

New repository

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk (*).

Owner * Repository name *

Tuchsanai / test_week1

test_week1 is available.

Great repository names are short and memorable. Need inspiration? How about [super-duper-disco](#)?

Description (optional)

Public Anyone on the internet can see this repository. You choose who can commit.

Private You choose who can see and commit to this repository.

Initialize this repository with:

☒ Add a README file This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: Python

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

License: None

A license tells others what they can and can't do with your code. [Learn more about licenses.](#)

This will set `main` as the default branch. Change the default name in your [settings](#).

ⓘ You are creating a private repository in your personal account.

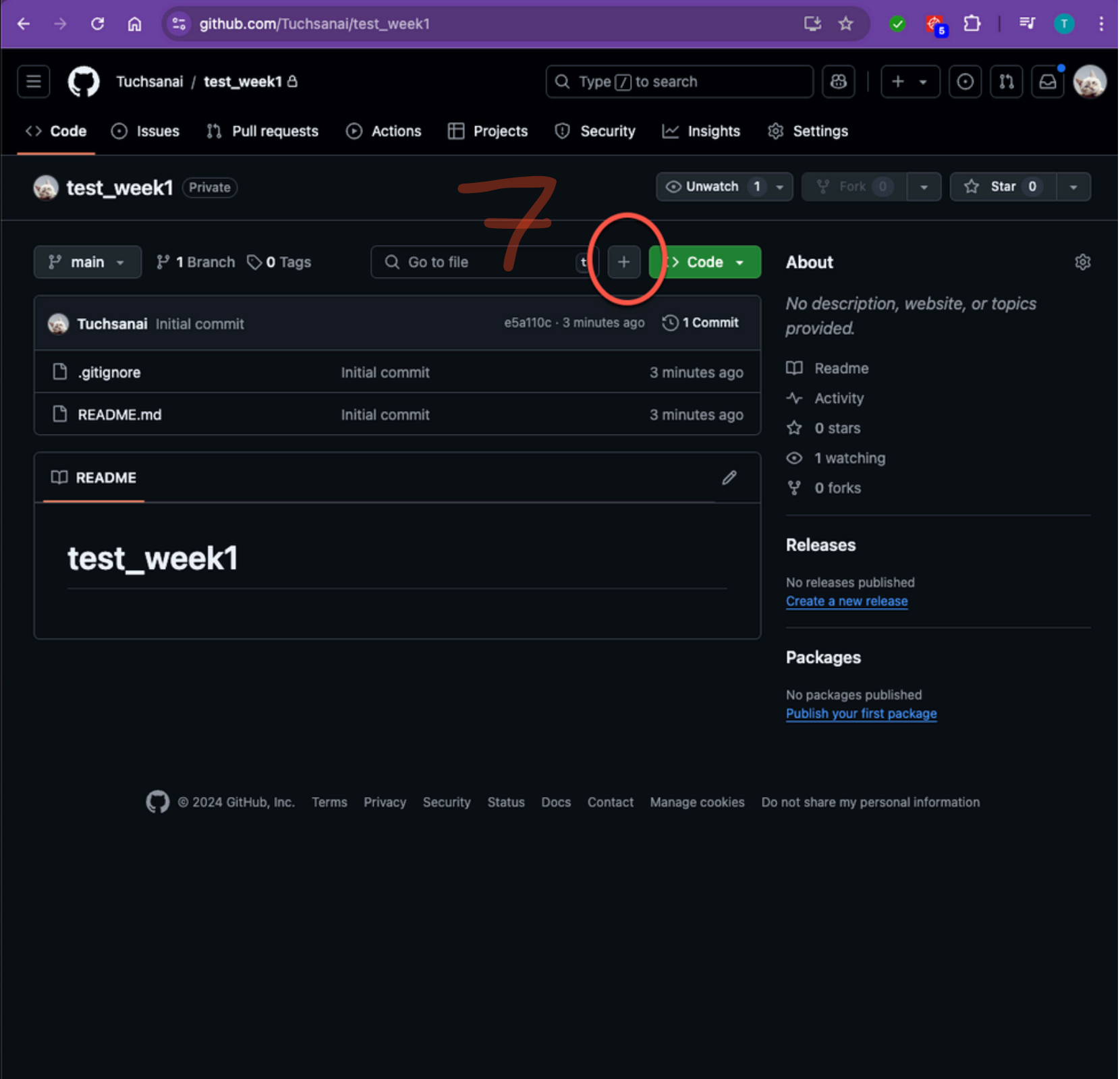
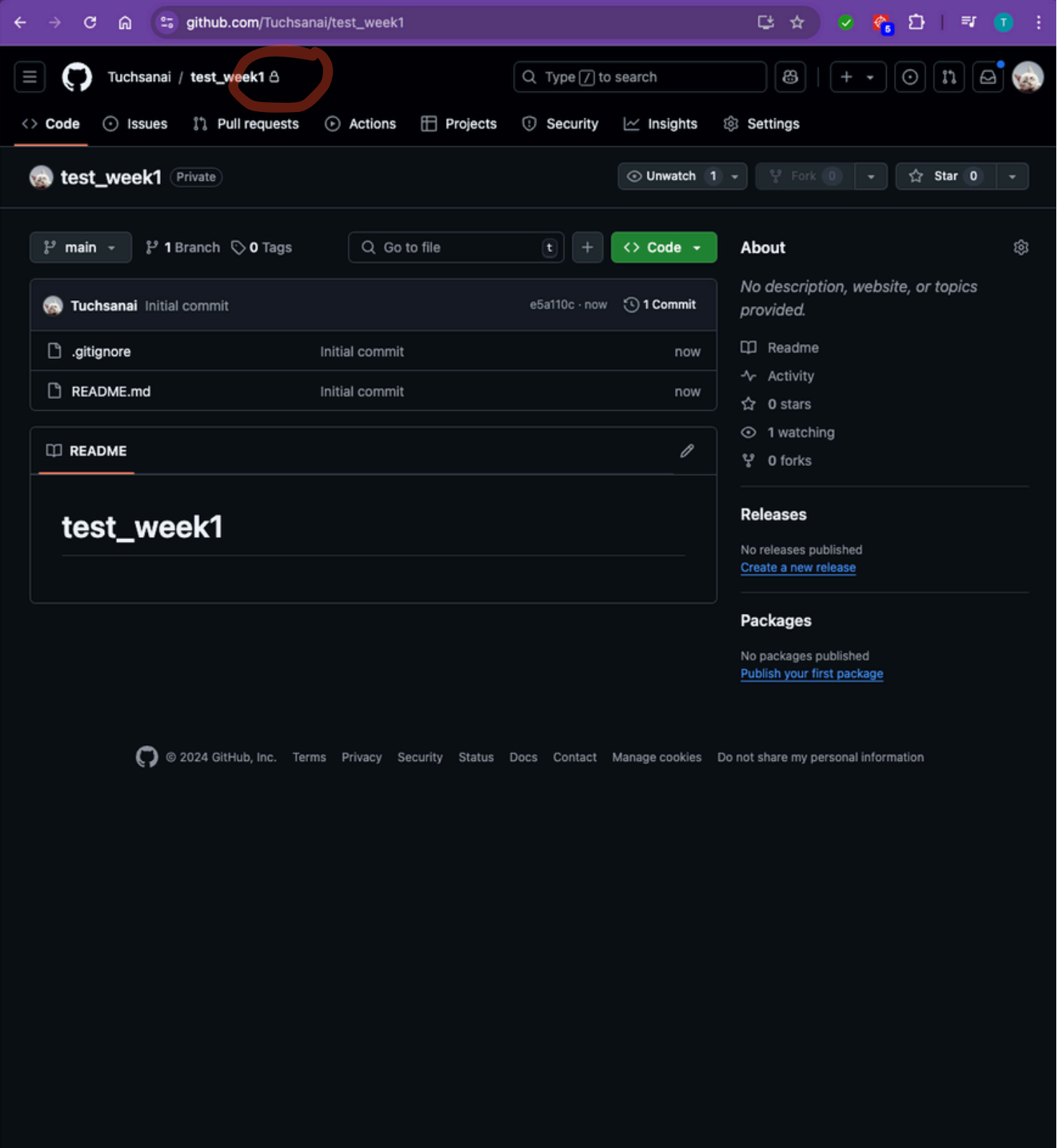
Create repository

4

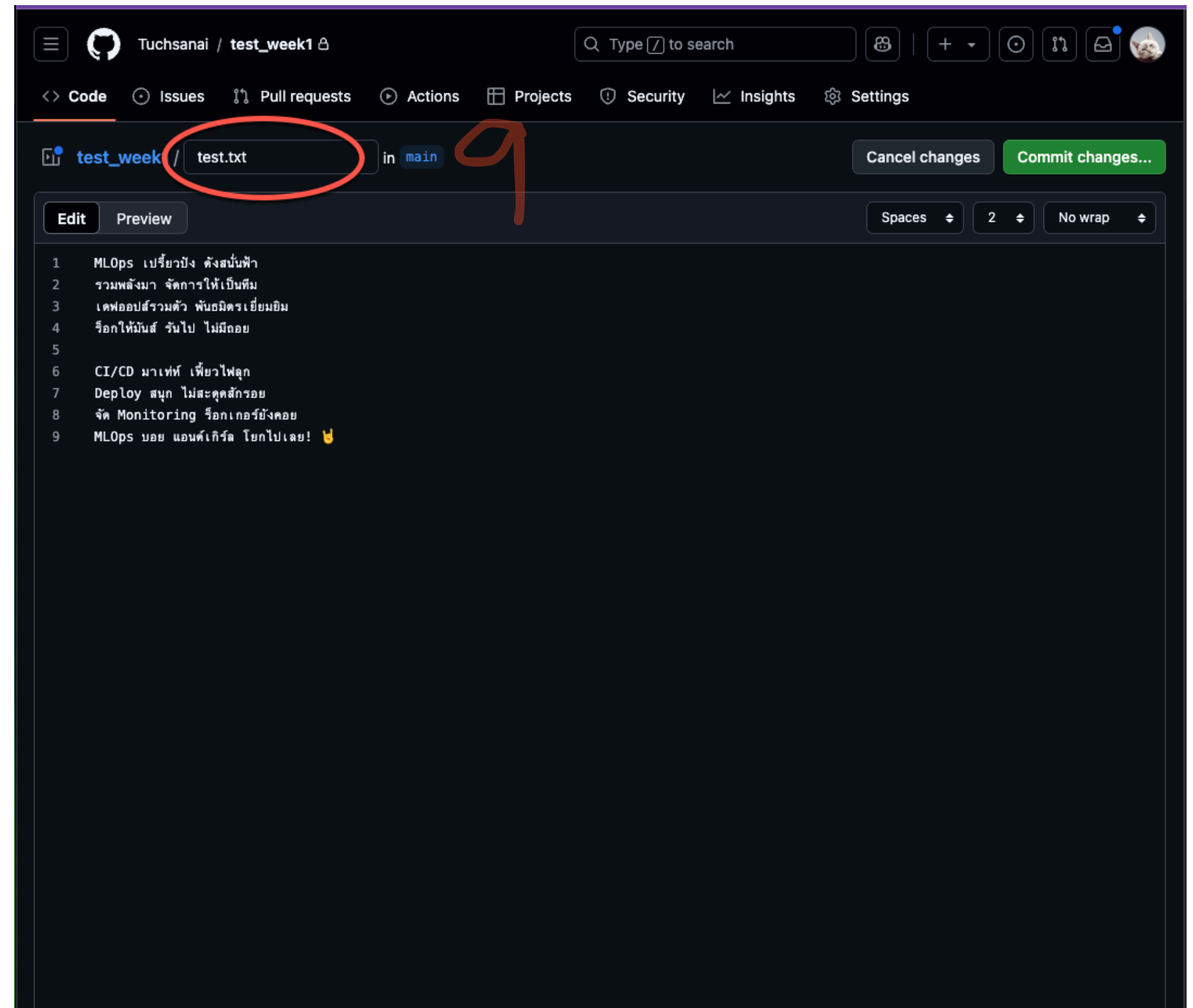
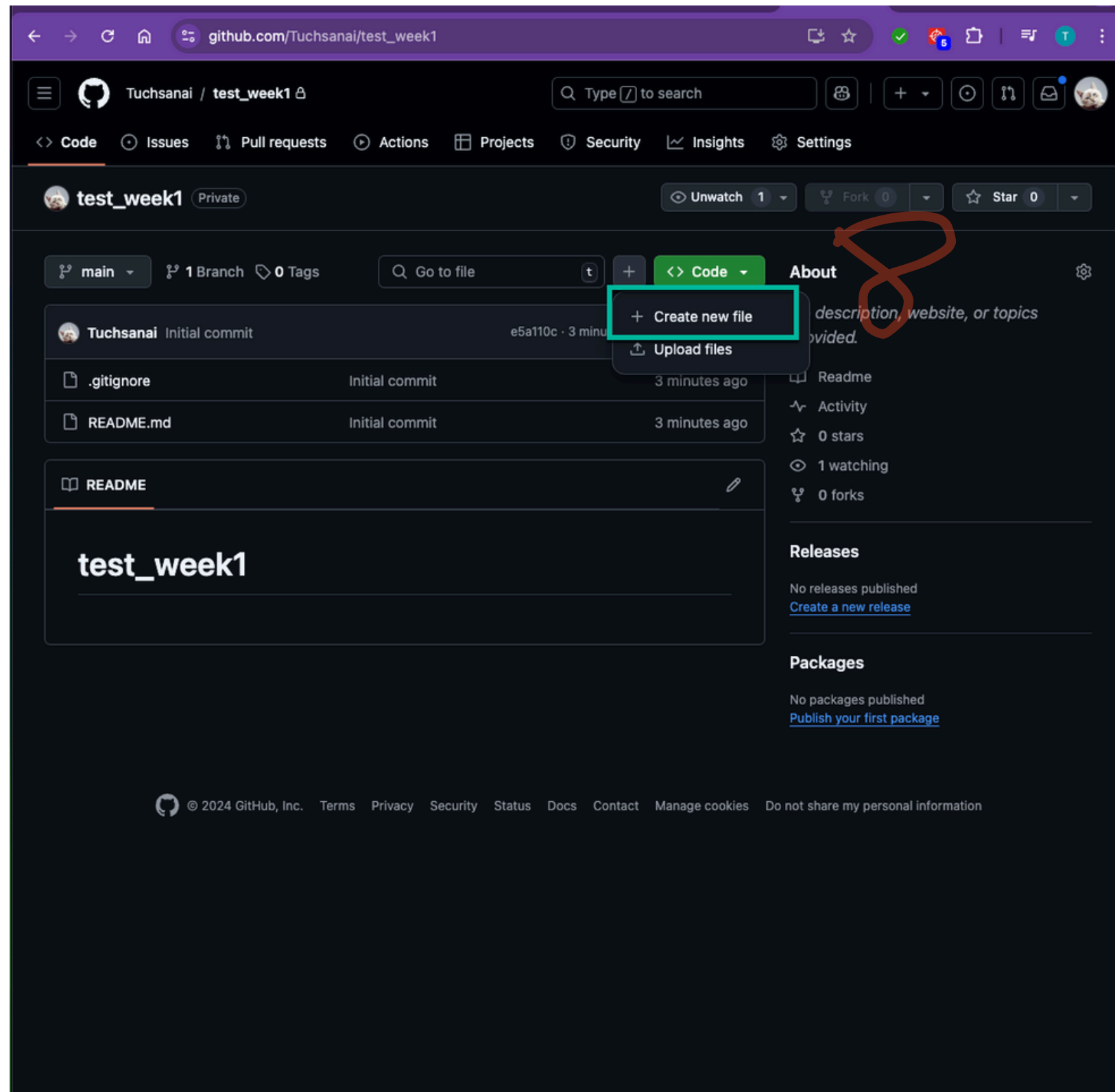
5

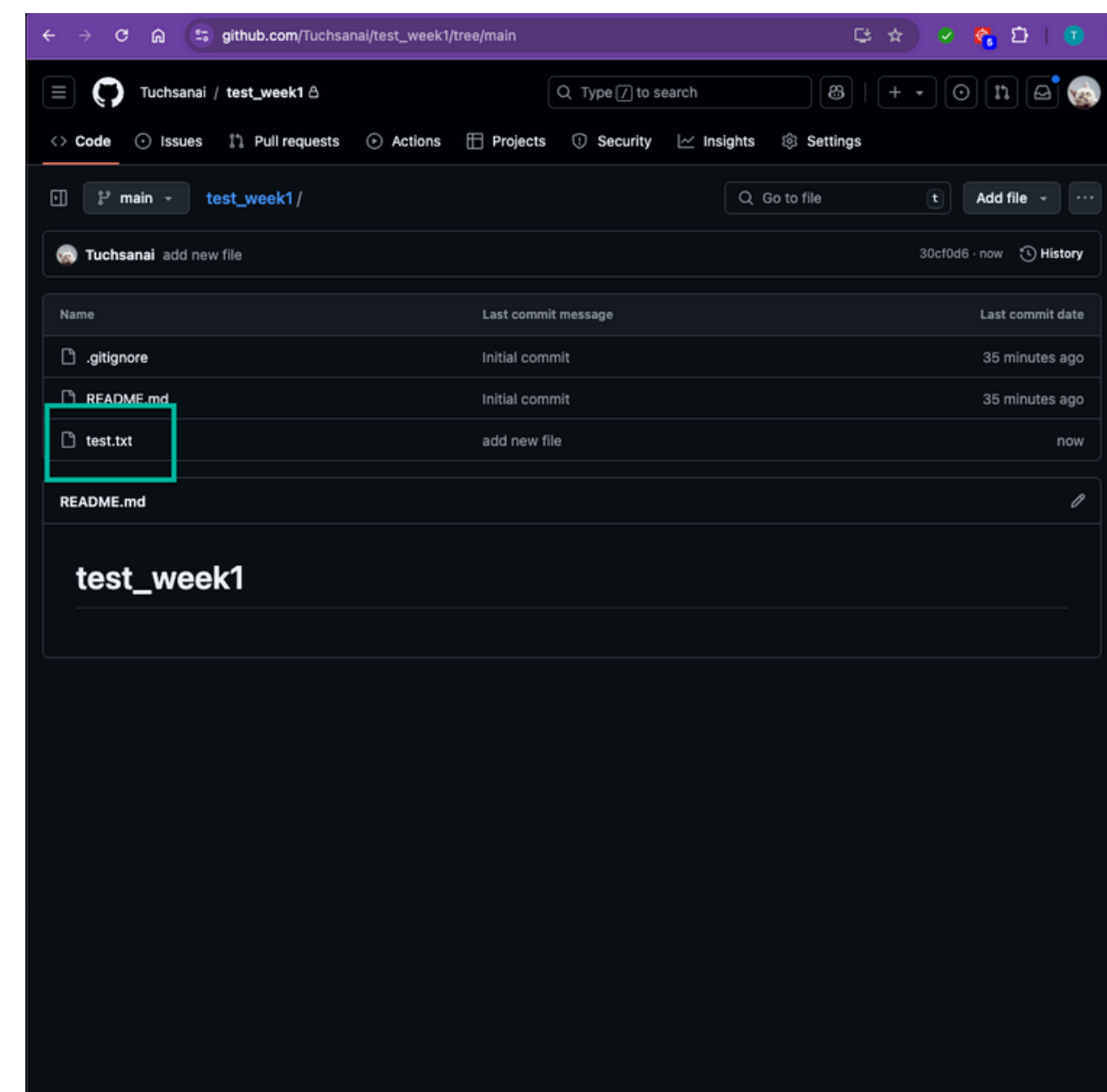
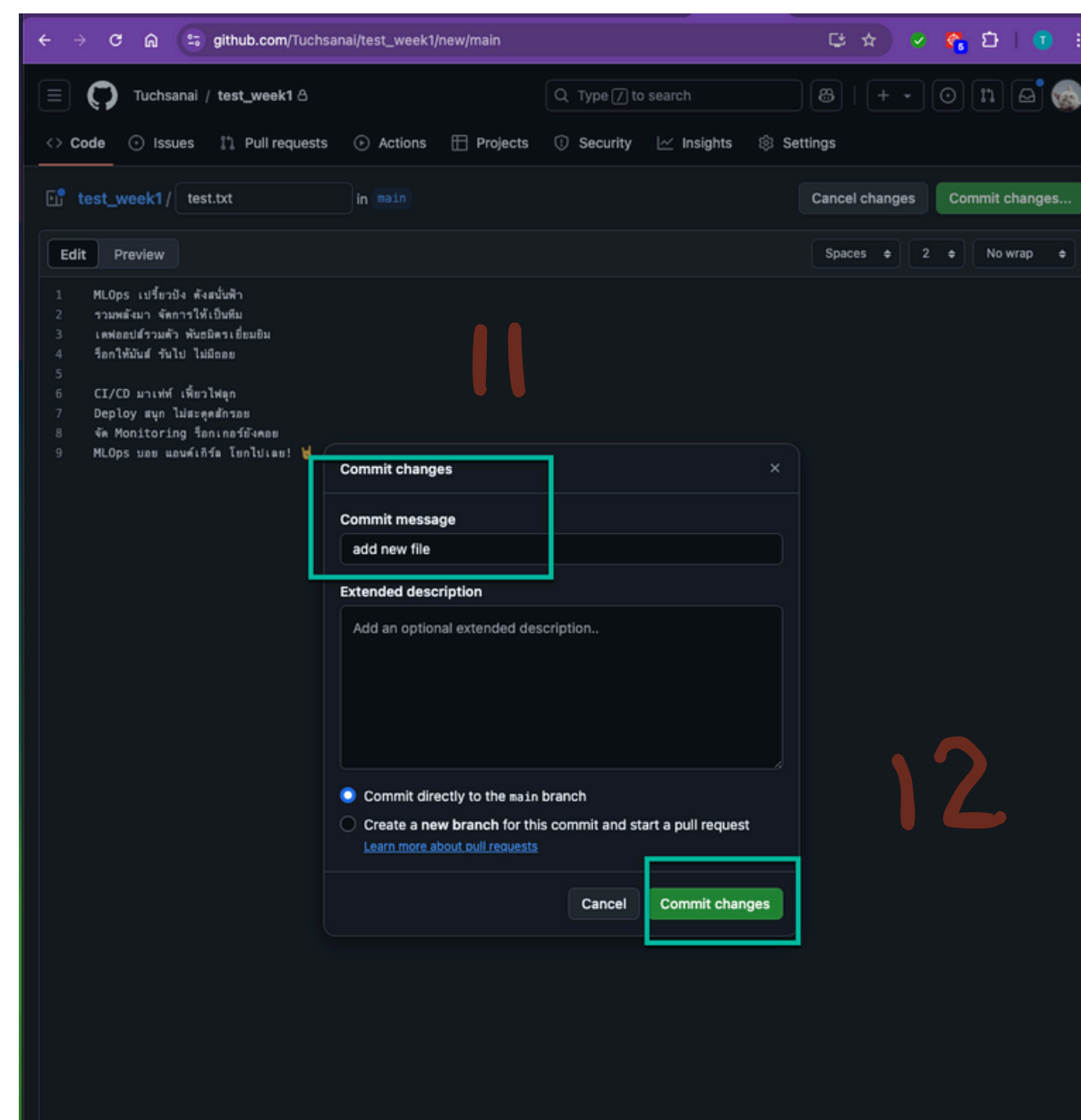
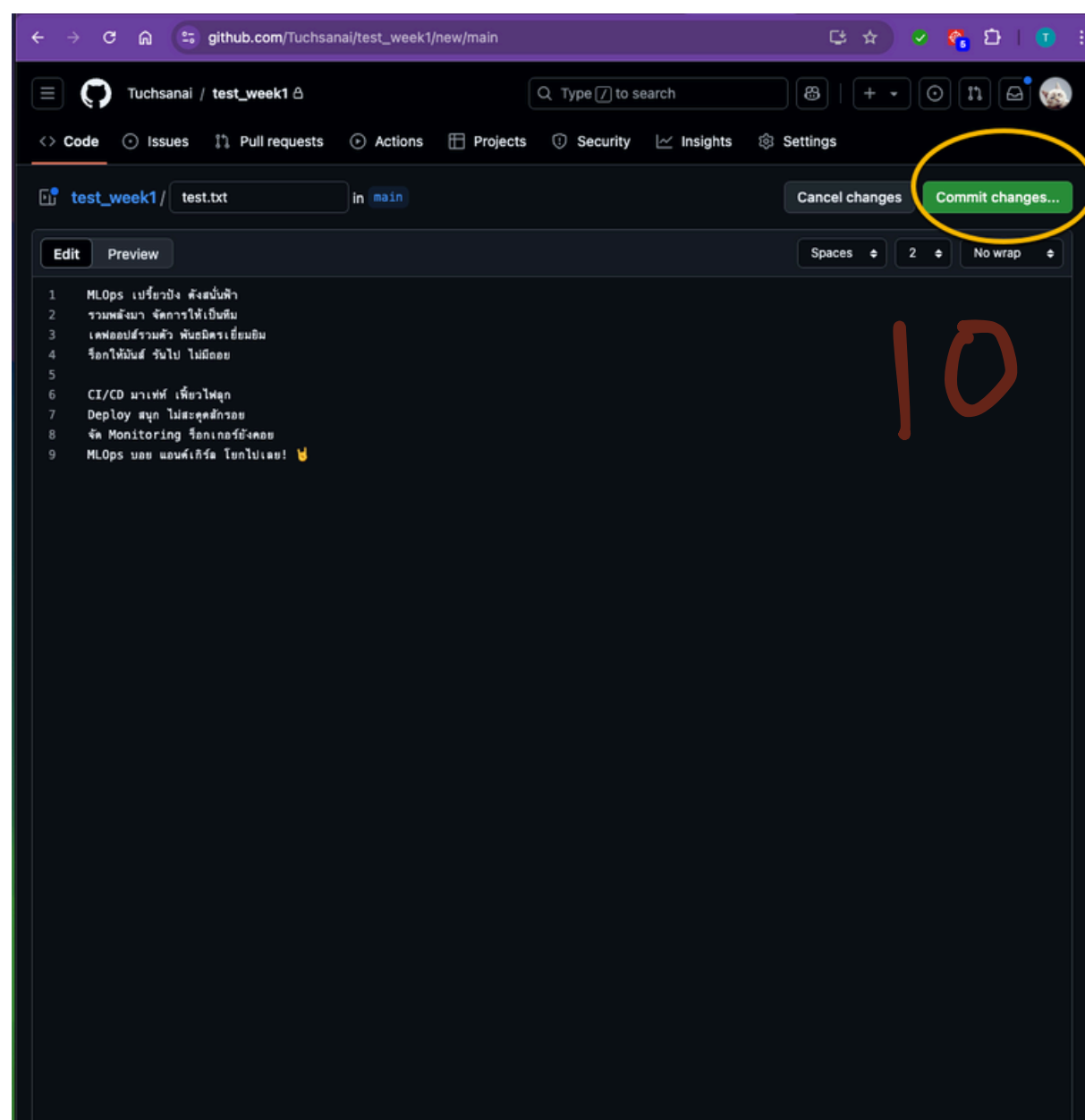
6

Privated repository



สร้าง file : test.txt





- Clone private repository:

git clone https://username:YOUR_TOKEN@github.com/username/repo.git